

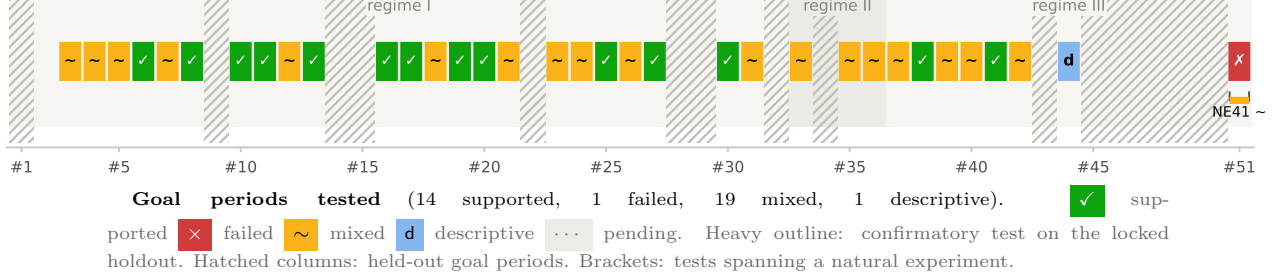
H73: Style = weights + context + register

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

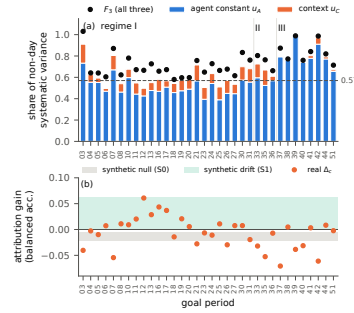
Bottom line. Style is an agent constant plus noise: context fill and assigned registers add small shares, and removing them does not improve attribution.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

The test splits an agent's chat style into an agent constant, a context-fill drift and an assigned-register shift. Style is H13's 17 type-controlled features. The fit runs per goal period on 34 non-holdout periods (126k messages). The three components explain most of the systematic variance that the day field leaves ($F_3 = 0.72$, IQR 0.65–0.80; 34/34 periods above 0.57). The agent constant carries almost all of it. Context fill adds a median share of 0.08, and 0.011 in regime III. Removing the context term does not raise attribution ($\Delta_c = -0.007$ [−0.017, +0.003]). At forced erasures the fitted drift predicts 1.7% of the style jump. Judging in #12 is a shared register. The #51 roles share none. The holdout script exists and is not run.



Observables. (a) Shares of the non-day systematic style variance per goal period. Blue is the agent constant, orange is the context term, and black dots are all three components (F_3). The agent constant is the largest share in every period. The context share falls to about 0.01 in regime III. (b) Attribution gain from the context model on blocks of five messages. Real gains (dots) sit in the synthetic null band, not in the planted-drift band.

PHYSICAL MODEL

Vector spins (model 11). The spin is a message's style vector x_m . The day field g_d and a time-of-day term τ are external fields. The agent constant q_i is the weights. The context term is a profile $c(s)$ over fill bins plus an agent drift b_i along $z = \log_2(1 + p)$, with p the calls since the last reset. ρ_r is an assigned register (judge, role). η is agent-day jitter. κ is the cell-means ceiling.

$$x_m = g_d + \tau(h_m) + q_i + c(s_m) + b_i z_m + \rho_{r_m} + \eta_{i,d} + \varepsilon_m$$

$$F_3 = \frac{\bar{R}^2(G+A+C+R) - \bar{R}^2(G)}{\kappa - \bar{R}^2(G)}, \quad u_X = \frac{\bar{R}^2(\text{all}) - \bar{R}^2(\text{all} \setminus X)}{\kappa - \bar{R}^2(G)}$$

$$\Delta_c = \text{acc}[q_j + c(s) + b_j z] - \text{acc}[q_j], \quad \beta = \frac{\sum \langle \Delta x, \Delta \hat{x}_{\text{ctx}} \rangle}{\sum \|\Delta \hat{x}_{\text{ctx}}\|^2}$$

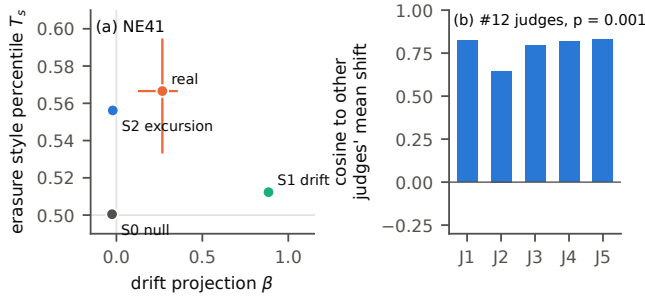
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For attribution, use the agent's mean style over at least five messages. Balanced accuracy is 0.63 against chance 0.10. Do not correct for context position: the correction gains −0.007. A style monitor can treat the agent constant as fixed within a period. It carries 0.55 of the non-day systematic variance, and agent-day jitter carries 0.18. Expect one message to be mostly noise: only 27% of its style variance is systematic. Known speech roles such as judging shift style in one shared direction ($\cos 0.78$). That shift does not confuse attribution. Context erasure moves style by a random amount with no fixed direction, so a monitor should not read a post-reset style change as a model change.

H73-style-three-components · generated 2026-10-04 12:49 by infra/summaries/build_summaries.py · card hypotheses/H73-style-three-components/README.md · ratings: H73 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $F_3 \geq 0.57$ in $\geq 2/3$	34/34; median 0.72 (agent-day ceiling 0.82)	pass
P2 agent largest; context $p < .05$	u_A largest 34/34 (0.55); u_C 23/26 (0.08)	pass
P3 context removal helps attribution	$\Delta_c -0.007 [-0.017, +0.003]$; 16/34 positive	fail
P4 dispersion grows with fill	1/5 dense periods; G51 negative	fail
P5 rivals (genre, own fill)	genre retention 1.04; received fill beats own fill 6/8 (regime III)	partial
N1 NE41 drift predicts the jump	β 0.27 [0.12, 0.36]; 1.7% of the jump; T_s 0.567 $\rightarrow$ 0.568	mixed
N2 #12 judges share a register	cos +0.78 (p 0.001); u_R 0.056; attribution +0.01	mixed
N3 #51 roles are registers	Prankster 0.60; media cos -0.11 (p 0.56)	fail
N4 leader writes like Kimi	rank 3 blind, 1 with agent drift (20 msgs)	descr.



Second figure. (a) Forced erasures (NE41): drift projection β against the style percentile T_s . The real point (orange, 95% CIs) sits near the synthetic directionless excursion (S2), far from a directed drift (S1). (b) #12: each judge's style shift points along the other judges' mean shift.

GOAL PERIODS IN DETAIL

G03 $\sim$ mixed
G04 $\sim$ mixed
G05 $\sim$ mixed
G06 $\checkmark$ supported
G07 $\sim$ mixed
G08 $\checkmark$ supported
G10 $\checkmark$ supported
G11 $\checkmark$ supported
G12 $\sim$ mixed
G13 $\checkmark$ supported

+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The 20 features mix formatting with topic-adjacent rates (digits, uppercase). F_3 rests on a cell ceiling that is noisy where cells are small. The synthetic bias reaches +0.07 near 0.55, and F_3 exceeds 1 in G03. In regimes I/II half of the context share is the chat versus computer-use level. Fill counts calls, not messages or tokens, and it may miss what drives the erasure excursion. The #51 role test is pre-registered on three-week eras. The onset-block result (Prankster 1.00) is post hoc. The natives are small: five judges, four media agents, 20 leader messages.

ROUND 2

Model the conversation state (received items, pending mentions, thread depth) in place of fill, with an in-flight placebo. Fit the persona shift as a transient with a decay time. Run the NE30 model swap on the holdout.

- **Essence.** which parts of an agent's style a monitor can trust as identity, and which parts the conversation writes.
- **H73-R1** Conversation-state model: replace fill with received items, pending @-mentions and thread depth (DQ2), and test accommodation with H57's in-flight placebo (posted-but-unread at matched lag).
- **H73-R2** Register as a transient: a daily event study of the #51 onset and NE38 (decay time of the persona shift), and the #12 judge shift by minute within a debate.
- **H73-R3** Excursion model: an Ornstein–Uhlenbeck process within a context segment on message position (H46-R2), with its direction random per segment; fit its variance growth and test it at NE41.
- **H73-R4** Model swaps: the NE30 test in the holdout (frozen as C6); in round 2 a larger leader sample (#45 is held out).